Class 14: RNA-Seq analysis mini-project

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Table of contents

Background . . . 1
Data Import . . . 1
    Check and Tidy . . . 4
Setup for DESeq . . . 5
Remove zero count genes . . . 6
    Run DESeq . . . 6
    Get results . . . 7
    Save results . . . 8
Volcano Plot . . . 8
Add gene annotation . . . 9
    MapIds . . . 9
    Save annotated results . . . 11
Pathway analysis . . . 11
GO analysis . . . 18
Reactome Analysis . . . 20

Background

> Trapnell C, Hendrickson DG, Sauvageau M, Goff L et al. "Differential analysis of gene regulation at transcript resolution with RNA-seq".

The authors report on differential analysis of lung fibroblasts in response to loss of the developmental transcription factor HOXA1.

Data Import

We have 2 key input files: counts and metadata.

```r
library(DESeq2)
```

```
Loading required package: S4Vectors

Loading required package: stats4

Loading required package: BiocGenerics

Loading required package: generics


Attaching package: 'generics'

The following objects are masked from 'package:base':

    as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
    setequal, union


Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

    IQR, mad, sd, var, xtabs

The following objects are masked from 'package:base':

    anyDuplicated, aperm, append, as.data.frame, basename, cbind,
    colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
    get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
    order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
    rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
    unsplit, which.max, which.min


Attaching package: 'S4Vectors'
```

```
The following object is masked from 'package:utils':

    findMatches

The following objects are masked from 'package:base':

    expand.grid, I, unname

Loading required package: IRanges

Loading required package: GenomicRanges

Loading required package: Seqinfo

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats


Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

    colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
    colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
    colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
    colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
    colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
    colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
    colWeightedMeans, colWeightedMedians, colWeightedSds,
    colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
    rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
    rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
    rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
    rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
    rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
    rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
    rowWeightedSds, rowWeightedVars
```

```
Loading required package: Biobase

Welcome to Bioconductor

    Vignettes contain introductory material; view with
    'browseVignettes()'. To cite Bioconductor, see
    'citation("Biobase")', and for packages 'citation("pkgname")'.


Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

    rowMedians

The following objects are masked from 'package:matrixStats':

    anyMissing, rowMedians
```

```r
metaFile <- "GSE37704_metadata.csv"
countFile <- "GSE37704_featurecounts.csv"
```

Check and Tidy

```r
# Import metadata and take a peek
colData <- read.csv(metaFile, row.names=1)
head(colData)
```

```
              condition
SRR493366 control_sirna
SRR493367 control_sirna
SRR493368 control_sirna
SRR493369      hoxa1_kd
SRR493370      hoxa1_kd
SRR493371      hoxa1_kd
```

```r
# Import countdata
countData <-read.csv(countFile, row.names=1)
head(countData)
```

```
                 length SRR493366 SRR493367 SRR493368 SRR493369 SRR493370
ENSG00000186092    918         0         0         0         0         0
ENSG00000279928    718         0         0         0         0         0
ENSG00000279457   1982        23        28        29        29        28
ENSG00000278566    939         0         0         0         0         0
ENSG00000273547    939         0         0         0         0         0
ENSG00000187634   3214       124       123       205       207       212
                 SRR493371
ENSG00000186092         0
ENSG00000279928         0
ENSG00000279457        46
ENSG00000278566         0
ENSG00000273547         0
ENSG00000187634       258
```

> Q. Complete the code below to remove the troublesome first column from countData.

We need to remove the first "length" column from `countData` to have a 1:1 correspondance with `colData` rows.

```r
countData <- as.matrix(countData[,-1])
head(countData)
```

```
                 SRR493366 SRR493367 SRR493368 SRR493369 SRR493370 SRR493371
ENSG00000186092         0         0         0         0         0         0
ENSG00000279928         0         0         0         0         0         0
ENSG00000279457        23        28        29        29        28        46
ENSG00000278566         0         0         0         0         0         0
ENSG00000273547         0         0         0         0         0         0
ENSG00000187634       124       123       205       207       212       258
```

Setup for DESeq

> Q. Complete the code below to filter countData to exclude genes (i.e. rows) where we have 0 read count across all samples (i.e. columns).

```r
dds <- DESeqDataSetFromMatrix(countData = countData,
                              colData = colData,
                              design = ~condition)
```

```
Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
design formula are characters, converting to factors
```

```r
head(dds)
```

```
class: DESeqDataSet
dim: 6 6
metadata(1): version
assays(1): counts
rownames(6): ENSG00000186092 ENSG00000279928 ... ENSG00000273547
  ENSG00000187634
rowData names(0):
colnames(6): SRR493366 SRR493367 ... SRR493370 SRR493371
colData names(1): condition
```

Remove zero count genes

Some genes (rows) have no count data (i.e. zero values). We should remove these before any further analysis.

```r
rownames(colData) == colnames(countData)
```

```
[1] TRUE TRUE TRUE TRUE TRUE TRUE
```

Run DESeq

```r
dds
```

```
class: DESeqDataSet
dim: 19808 6
metadata(1): version
assays(1): counts
rownames(19808): ENSG00000186092 ENSG00000279928 ... ENSG00000277475
  ENSG00000268674
rowData names(0):
colnames(6): SRR493366 SRR493367 ... SRR493370 SRR493371
colData names(1): condition
```

Get results

```r
dds = DESeq(dds)
```

```
estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing
```

```r
res <- results(dds)
head(res)
```

```
log2 fold change (MLE): condition hoxa1 kd vs control sirna
Wald test p-value: condition hoxa1 kd vs control sirna
DataFrame with 6 rows and 6 columns
                 baseMean log2FoldChange     lfcSE      stat     pvalue
                <numeric>      <numeric> <numeric> <numeric>  <numeric>
ENSG00000186092    0.0000             NA        NA        NA         NA
ENSG00000279928    0.0000             NA        NA        NA         NA
ENSG00000279457   29.9136       0.179257  0.324822  0.551863 0.58104205
ENSG00000278566    0.0000             NA        NA        NA         NA
ENSG00000273547    0.0000             NA        NA        NA         NA
ENSG00000187634  183.2296       0.426457  0.140266  3.040350 0.00236304
                     padj
                 <numeric>
ENSG00000186092         NA
ENSG00000279928         NA
ENSG00000279457 0.68707978
ENSG00000278566         NA
ENSG00000273547         NA
ENSG00000187634 0.00516278
```

Save results

```
write.csv(res, file="results.csv")
```

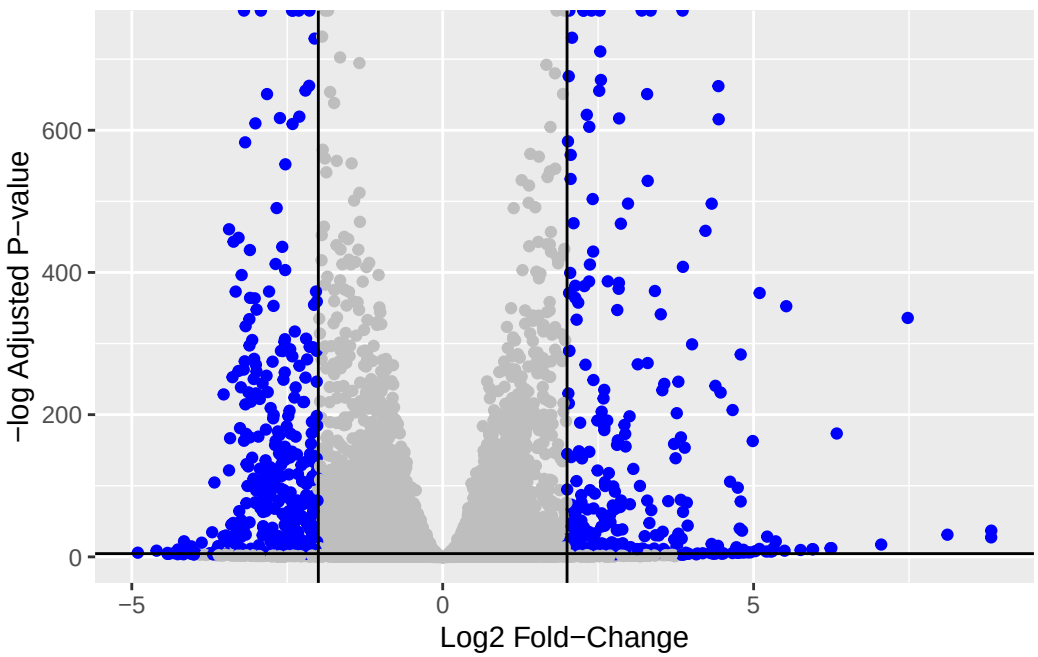
Volcano Plot

```
library(ggplot2)

mycols <- rep("gray", nrow(res))
mycols[res$log2FoldChange > 2] <- "blue"
mycols[res$log2FoldChange < -2] <- "blue"
mycols[res$padj >= 0.05 ] <- "gray"

ggplot(res) +
  aes(log2FoldChange,
      -log(padj)) +
 geom_point(color=mycols) +
labs(x="Log2 Fold-Change", y="-log Adjusted P-value") +
geom_vline(xintercept = c(-2,2)) +
geom_hline(yintercept = -log(0.01))
```

```
Warning: Removed 5054 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Add gene annotation

```
library("AnnotationDbi")
library("org.Hs.eg.db")
```

```
columns(org.Hs.eg.db)
```

```
 [1] "ACCNUM"       "ALIAS"        "ENSEMBL"      "ENSEMBLPROT"  "ENSEMBLTRANS"
 [6] "ENTREZID"     "ENZYME"       "EVIDENCE"     "EVIDENCEALL"  "GENENAME"
[11] "GENETYPE"     "GO"           "GOALL"        "IPI"          "MAP"
[16] "OMIM"         "ONTOLOGY"     "ONTOLOGYALL"  "PATH"         "PFAM"
[21] "PMID"         "PROSITE"      "REFSEQ"       "SYMBOL"       "UCSCKG"
[26] "UNIPROT"
```

MapIds

```r
columns(org.Hs.eg.db)
```

```
 [1] "ACCNUM"       "ALIAS"        "ENSEMBL"      "ENSEMBLPROT"  "ENSEMBLTRANS"
 [6] "ENTREZID"     "ENZYME"       "EVIDENCE"     "EVIDENCEALL"  "GENENAME"
[11] "GENETYPE"     "GO"           "GOALL"        "IPI"          "MAP"
[16] "OMIM"         "ONTOLOGY"     "ONTOLOGYALL"  "PATH"         "PFAM"
[21] "PMID"         "PROSITE"      "REFSEQ"       "SYMBOL"       "UCSCKG"
[26] "UNIPROT"
```

```r
res$symbol = mapIds(org.Hs.eg.db,
                    keys=row.names(res),
                    keytype="ENSEMBL",
                    column="SYMBOL")
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
res$entrez = mapIds(org.Hs.eg.db,
                    keys=row.names(res),
                    keytype="ENSEMBL",
                    column="ENTREZID")
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
res$name =   mapIds(org.Hs.eg.db,
                    keys=row.names(res),
                    keytype="ENSEMBL",
                    column="GENENAME")
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
head(res, 10)
```

```
log2 fold change (MLE): condition hoxa1 kd vs control sirna
Wald test p-value: condition hoxa1 kd vs control sirna
DataFrame with 10 rows and 9 columns
                 baseMean log2FoldChange     lfcSE       stat      pvalue
                <numeric>      <numeric> <numeric>  <numeric>   <numeric>
ENSG00000186092    0.0000             NA        NA         NA          NA
```

```
ENSG00000279928    0.0000             NA        NA         NA          NA
ENSG00000279457   29.9136      0.1792571 0.3248216   0.551863 5.81042e-01
ENSG00000278566    0.0000             NA        NA         NA          NA
ENSG00000273547    0.0000             NA        NA         NA          NA
ENSG00000187634  183.2296      0.4264571 0.1402658   3.040350 2.36304e-03
ENSG00000188976 1651.1881     -0.6927205 0.0548465 -12.630158 1.43990e-36
ENSG00000187961  209.6379      0.7297556 0.1318599   5.534326 3.12428e-08
ENSG00000187583   47.2551      0.0405765 0.2718928   0.149237 8.81366e-01
ENSG00000187642   11.9798      0.5428105 0.5215598   1.040744 2.97994e-01
                     padj      symbol      entrez                   name
                <numeric> <character> <character>            <character>
ENSG00000186092          NA       OR4F5       79501 olfactory receptor f..
ENSG00000279928          NA          NA          NA                     NA
ENSG00000279457 6.87080e-01          NA          NA                     NA
ENSG00000278566          NA          NA          NA                     NA
ENSG00000273547          NA          NA          NA                     NA
ENSG00000187634 5.16278e-03      SAMD11      148398 sterile alpha motif ..
ENSG00000188976 1.76741e-35       NOC2L       26155 NOC2 like nucleolar ..
ENSG00000187961 1.13536e-07      KLHL17      339451 kelch like family me..
ENSG00000187583 9.18988e-01     PLEKHN1       84069 pleckstrin homology ..
ENSG00000187642 4.03817e-01       PERM1       84808 PPARGC1 and ESRR ind..
```

Save annotated results

```r
write.csv(res, file="results_annotated.csv")
```

Pathway analysis

```r
library(pathview)
library(gage)
library(gageData)
```

```r
data(kegg.sets.hs)
data(sigmet.idx.hs)

# Focus on signaling and metabolic pathways only
kegg.sets.hs = kegg.sets.hs[sigmet.idx.hs]
```

```r
# Examine the first 3 pathways
head(kegg.sets.hs, 3)
```

```
$`hsa00232 Caffeine metabolism`
[1] "10"   "1544" "1548" "1549" "1553" "7498" "9"

$`hsa00983 Drug metabolism - other enzymes`
 [1] "10"     "1066"   "10720"  "10941"  "151531" "1548"   "1549"   "1551"
 [9] "1553"   "1576"   "1577"   "1806"   "1807"   "1890"   "221223" "2990"
[17] "3251"   "3614"   "3615"   "3704"   "51733"  "54490"  "54575"  "54576"
[25] "54577"  "54578"  "54579"  "54600"  "54657"  "54658"  "54659"  "54963"
[33] "574537" "64816"  "7083"   "7084"   "7172"   "7363"   "7364"   "7365"
[41] "7366"   "7367"   "7371"   "7372"   "7378"   "7498"   "79799"  "83549"
[49] "8824"   "8833"   "9"      "978"

$`hsa00230 Purine metabolism`
  [1] "100"    "10201"  "10606"  "10621"  "10622"  "10623"  "107"    "10714"
  [9] "108"    "10846"  "109"    "111"    "11128"  "11164"  "112"    "113"
 [17] "114"    "115"    "122481" "122622" "124583" "132"    "158"    "159"
 [25] "1633"   "171568" "1716"   "196883" "203"    "204"    "205"    "221823"
 [33] "2272"   "22978"  "23649"  "246721" "25885"  "2618"   "26289"  "270"
 [41] "271"    "27115"  "272"    "2766"   "2977"   "2982"   "2983"   "2984"
 [49] "2986"   "2987"   "29922"  "3000"   "30833"  "30834"  "318"    "3251"
 [57] "353"    "3614"   "3615"   "3704"   "377841" "471"    "4830"   "4831"
 [65] "4832"   "4833"   "4860"   "4881"   "4882"   "4907"   "50484"  "50940"
 [73] "51082"  "51251"  "51292"  "5136"   "5137"   "5138"   "5139"   "5140"
 [81] "5141"   "5142"   "5143"   "5144"   "5145"   "5146"   "5147"   "5148"
 [89] "5149"   "5150"   "5151"   "5152"   "5153"   "5158"   "5167"   "5169"
 [97] "51728"  "5198"   "5236"   "5313"   "5315"   "53343"  "54107"  "5422"
[105] "5424"   "5425"   "5426"   "5427"   "5430"   "5431"   "5432"   "5433"
[113] "5434"   "5435"   "5436"   "5437"   "5438"   "5439"   "5440"   "5441"
[121] "5471"   "548644" "55276"  "5557"   "5558"   "55703"  "55811"  "55821"
[129] "5631"   "5634"   "56655"  "56953"  "56985"  "57804"  "58497"  "6240"
[137] "6241"   "64425"  "646625" "654364" "661"    "7498"   "8382"   "84172"
[145] "84265"  "84284"  "84618"  "8622"   "8654"   "87178"  "8833"   "9060"
[153] "9061"   "93034"  "953"    "9533"   "954"    "955"    "956"    "957"
[161] "9583"   "9615"
```

```r
foldchanges <- res$log2FoldChange
names(foldchanges) <- res$entrez
head(foldchanges)
```

```
    79501      <NA>      <NA>      <NA>      <NA>    148398
       NA        NA 0.1792571        NA        NA 0.4264571
```

```
names(foldchanges) <- res$entrez
```

```
keggres = gage(foldchanges, gsets=kegg.sets.hs)
attributes(keggres)
```

```
$names
[1] "greater" "less"    "stats"
```

```
head(keggres$less)
```

```
                                        p.geomean stat.mean        p.val
hsa04110 Cell cycle                   7.077982e-06 -4.432593 7.077982e-06
hsa03030 DNA replication              9.424076e-05 -3.951803 9.424076e-05
hsa03013 RNA transport                1.048017e-03 -3.112129 1.048017e-03
hsa04114 Oocyte meiosis               2.563806e-03 -2.827297 2.563806e-03
hsa03440 Homologous recombination     3.066756e-03 -2.852899 3.066756e-03
hsa00010 Glycolysis / Gluconeogenesis 4.360092e-03 -2.663825 4.360092e-03
                                            q.val set.size         exp1
hsa04110 Cell cycle                   0.001160789      124 7.077982e-06
hsa03030 DNA replication              0.007727742       36 9.424076e-05
hsa03013 RNA transport                0.057291598      149 1.048017e-03
hsa04114 Oocyte meiosis               0.100589607      112 2.563806e-03
hsa03440 Homologous recombination     0.100589607       28 3.066756e-03
hsa00010 Glycolysis / Gluconeogenesis 0.119175854       65 4.360092e-03
```

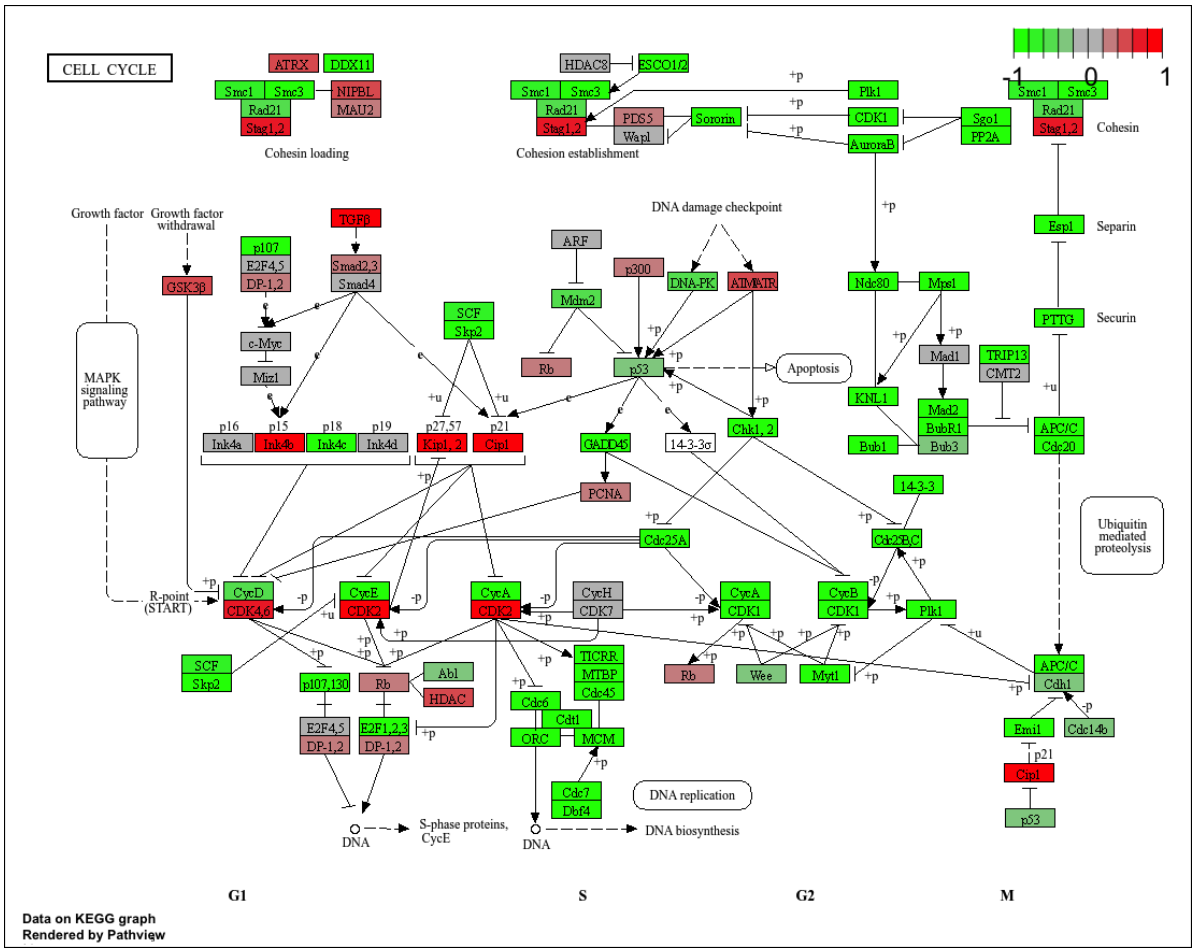
Cell cycle:

```
pathview(gene.data=foldchanges, pathway.id="hsa04110")
```

```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/dariana/BIMM143.w26/class14

Info: Writing image file hsa04110.pathview.png
```



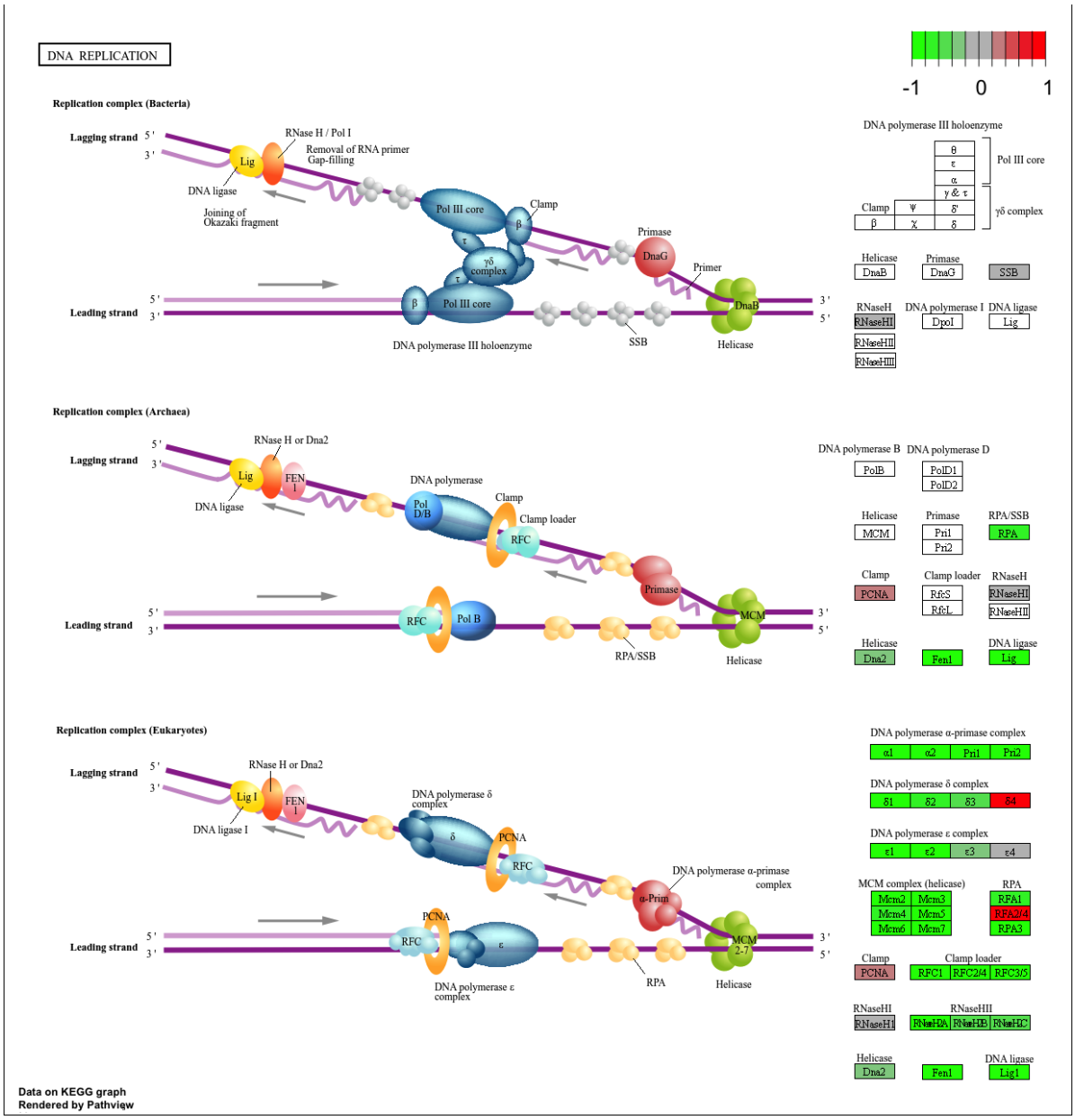
DNA replication:

```
pathview(gene.data=foldchanges, pathway.id="hsa03030")
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/dariana/BIMM143.w26/class14
```

```
Info: Writing image file hsa03030.pathview.png
```



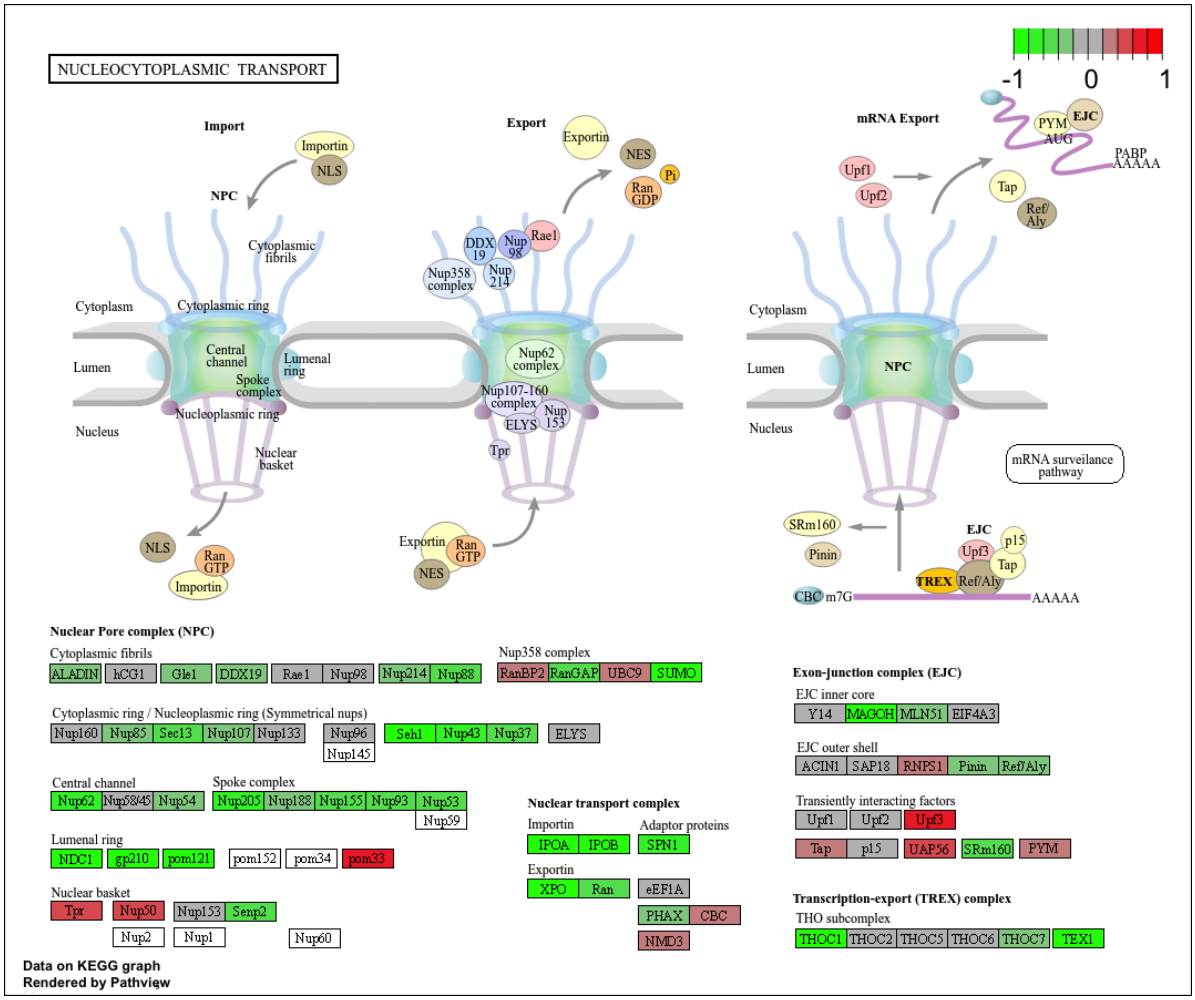
RNA transport:

```
pathview(gene.data=foldchanges, pathway.id="hsa03013")
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/dariana/BIMM143.w26/class14
```

```
Info: Writing image file hsa03013.pathview.png
```



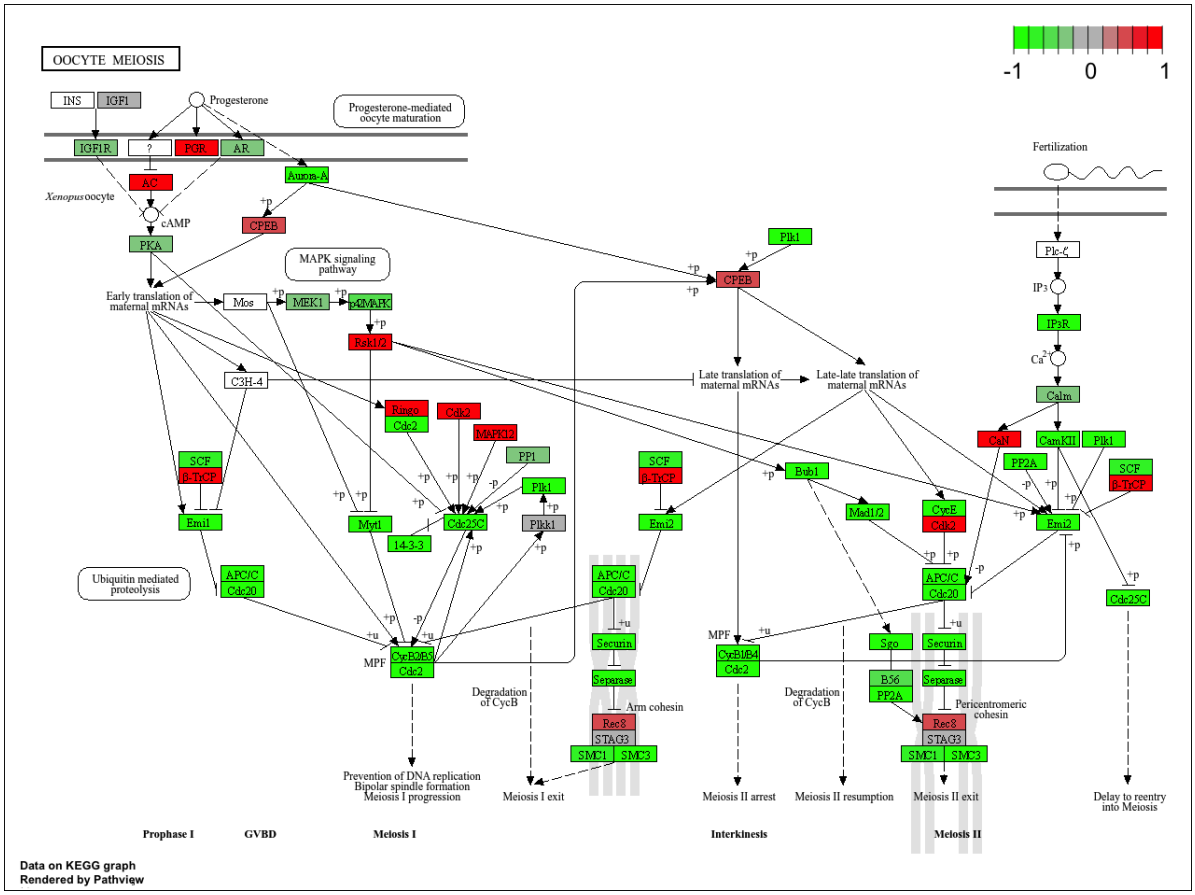
Oocyte meiosis:

```
pathview(gene.data=foldchanges, pathway.id="hsa04114")
```

```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/dariana/BIMM143.w26/class14

Info: Writing image file hsa04114.pathview.png
```



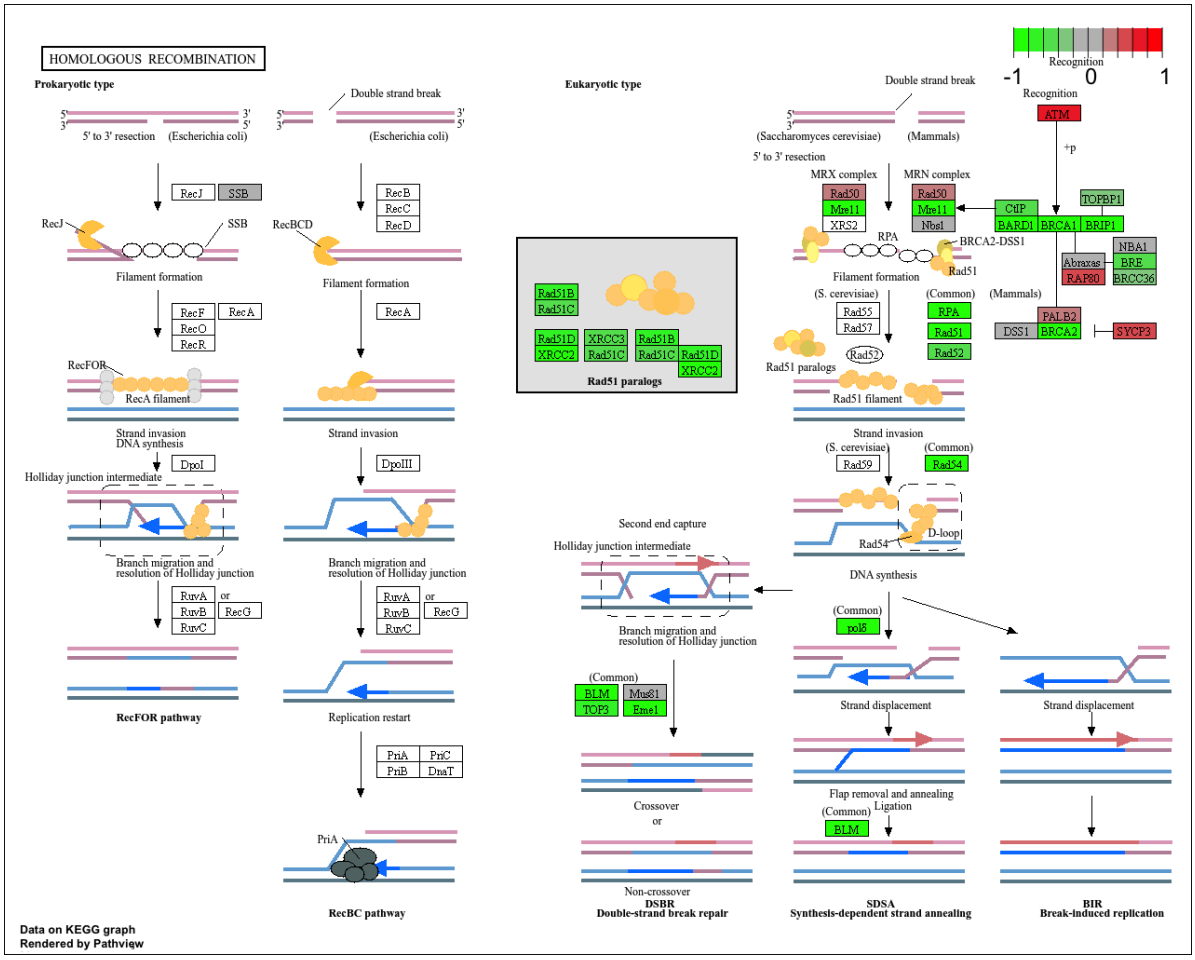
Homologous recombination:

```
pathview(gene.data=foldchanges, pathway.id="hsa03440")
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/dariana/BIMM143.w26/class14
```

```
Info: Writing image file hsa03440.pathview.png
```



GO analysis

Focus on hte biological process BP section

```
data(go.sets.hs)
data(go.subs.hs)

# Focus on Biological Process subset of GO
gobpsets = go.sets.hs[go.subs.hs$BP]

gobpres = gage(foldchanges, gsets=gobpsets)

lapply(gobpres, head)
```

```
$greater
                                            p.geomean stat.mean        p.val
GO:0007156 homophilic cell adhesion        1.734864e-05  4.210777 1.734864e-
05
GO:0048729 tissue morphogenesis            5.407952e-05  3.888470 5.407952e-
05
GO:0002009 morphogenesis of an epithelium  5.727599e-05  3.878706 5.727599e-
05
GO:0030855 epithelial cell differentiation 2.053700e-04  3.554776 2.053700e-
04
GO:0060562 epithelial tube morphogenesis   2.927804e-04  3.458463 2.927804e-
04
GO:0048598 embryonic morphogenesis         2.959270e-04  3.446527 2.959270e-
04
                                                 q.val set.size         exp1
GO:0007156 homophilic cell adhesion        0.07584825      137 1.734864e-05
GO:0048729 tissue morphogenesis            0.08347021      483 5.407952e-05
GO:0002009 morphogenesis of an epithelium  0.08347021      382 5.727599e-05
GO:0030855 epithelial cell differentiation 0.16449701      299 2.053700e-04
GO:0060562 epithelial tube morphogenesis   0.16449701      289 2.927804e-04
GO:0048598 embryonic morphogenesis         0.16449701      498 2.959270e-04

$less
                                             p.geomean stat.mean        p.val
GO:0048285 organelle fission             6.386337e-16 -8.175381 6.386337e-16
GO:0000280 nuclear division              1.726380e-15 -8.056666 1.726380e-15
GO:0007067 mitosis                       1.726380e-15 -8.056666 1.726380e-15
GO:0000087 M phase of mitotic cell cycle 4.593581e-15 -7.919909 4.593581e-15
GO:0007059 chromosome segregation        9.576332e-12 -6.994852 9.576332e-12
GO:0051301 cell division                 8.718528e-11 -6.455491 8.718528e-11
                                                 q.val set.size         exp1
GO:0048285 organelle fission             2.515911e-12      386 6.386337e-16
GO:0000280 nuclear division              2.515911e-12      362 1.726380e-15
GO:0007067 mitosis                       2.515911e-12      362 1.726380e-15
GO:0000087 M phase of mitotic cell cycle 5.020784e-12      373 4.593581e-15
GO:0007059 chromosome segregation        8.373545e-09      146 9.576332e-12
GO:0051301 cell division                 6.352901e-08      479 8.718528e-11

$stats
                                            stat.mean     exp1
GO:0007156 homophilic cell adhesion          4.210777 4.210777
GO:0048729 tissue morphogenesis              3.888470 3.888470
GO:0002009 morphogenesis of an epithelium    3.878706 3.878706
```

```
GO:0030855 epithelial cell differentiation  3.554776 3.554776
GO:0060562 epithelial tube morphogenesis    3.458463 3.458463
GO:0048598 embryonic morphogenesis          3.446527 3.446527
```

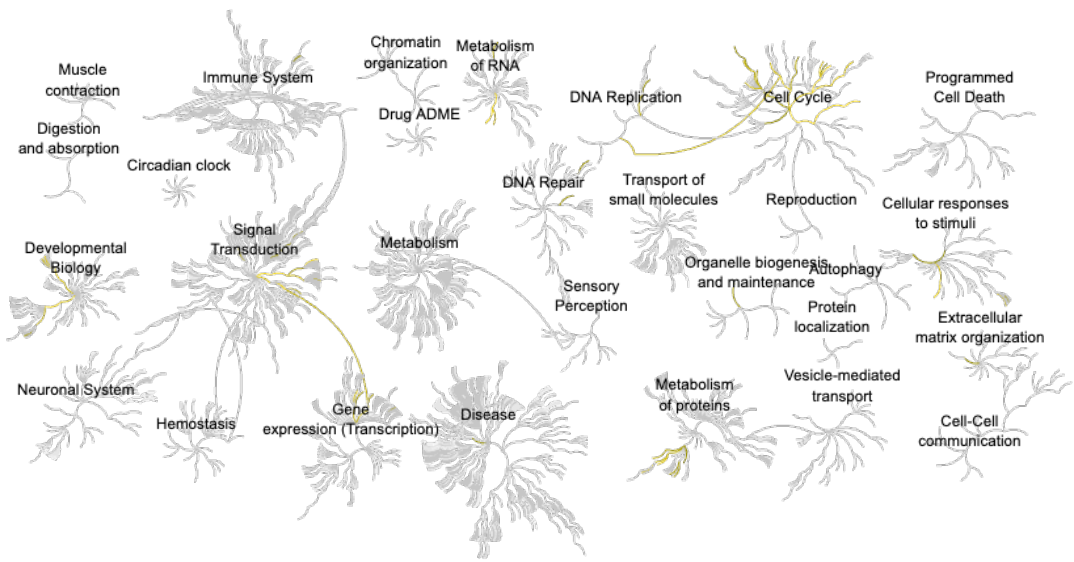
Reactome Analysis

We can use

```r
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]
print(paste("Total number of significant genes:", length(sig_genes)))
```

```
[1] "Total number of significant genes: 8146"
```

```r
write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=
```



>Q: What pathway has the most significant "Entities p-value"? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

Cell cycle, mitotic

yes most significant pathways match previous results